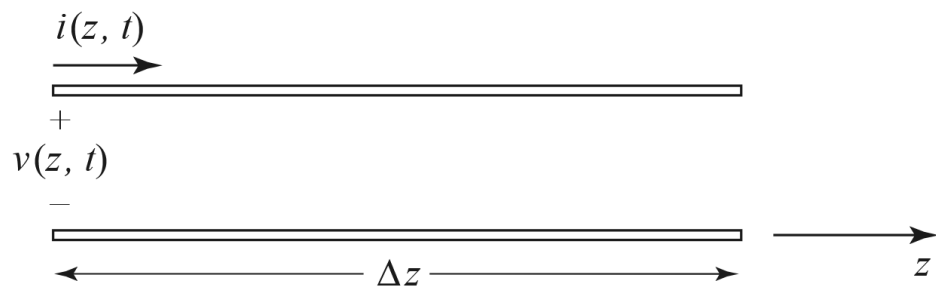


微波电路原理与设计

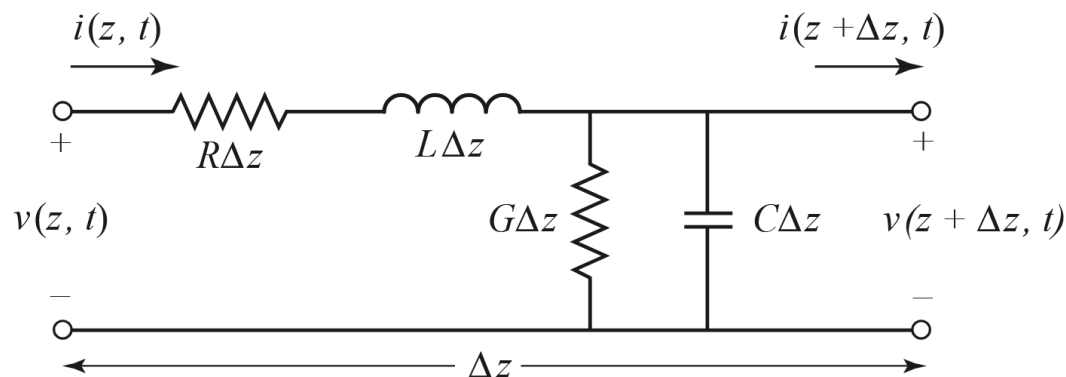
第一章传输线理论

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2026-03

传输线的物理模型



(a)



(b)

KVL

$$v(z, t) - R\Delta z i(z, t) - L\Delta z \frac{\partial i(z, t)}{\partial t} - v(z + \Delta z, t) = 0,$$

KCL

$$i(z, t) - G\Delta z v(z + \Delta z, t) - C\Delta z \frac{\partial v(z + \Delta z, t)}{\partial t} - i(z + \Delta z, t) = 0.$$

$\Delta z \rightarrow 0$ ↓

$$\frac{\partial v(z, t)}{\partial z} = -Ri(z, t) - L \frac{\partial i(z, t)}{\partial t},$$

$$\frac{\partial i(z, t)}{\partial z} = -Gv(z, t) - C \frac{\partial v(z, t)}{\partial t}.$$

↓

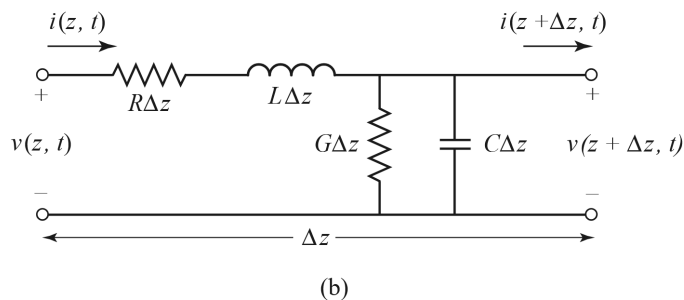
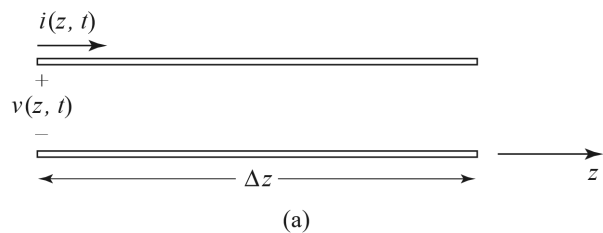
$$v(z, t) = \text{Re}\{V(z)e^{j\omega t}\}$$

$$i(z, t) = \text{Re}\{I(z)e^{j\omega t}\}$$

$$\frac{dV(z)}{dz} = -(R + j\omega L)I(z),$$

$$\frac{dI(z)}{dz} = -(G + j\omega C)V(z).$$

传输线模型



$$\frac{d^2 V(z)}{dz^2} - \gamma^2 V(z) = 0,$$

$$\frac{d^2 I(z)}{dz^2} - \gamma^2 I(z) = 0,$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

正向传输 反向传输

电压波 $V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z},$

电流波 $I(z) = I_o^+ e^{-\gamma z} + I_o^- e^{\gamma z},$

$$I(z) = \frac{\gamma}{R + j\omega L} (V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}). \quad Z_0 = \frac{R + j\omega L}{\gamma} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}, \quad \frac{V_o^+}{I_o^+} = Z_0 = \frac{-V_o^-}{I_o^-}. \quad I(z) = \frac{V_o^+}{Z_0} e^{-\gamma z} - \frac{V_o^-}{Z_0} e^{\gamma z}.$$

无耗传输线方程:

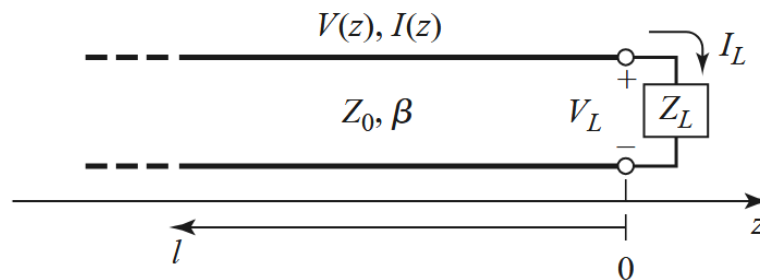
$$V(z) = V_o^+ e^{-j\beta z} + V_o^- e^{j\beta z},$$

$$I(z) = \frac{V_o^+}{Z_0} e^{-j\beta z} - \frac{V_o^-}{Z_0} e^{j\beta z}.$$

$$\beta = \omega \sqrt{LC} = \frac{2\pi}{\lambda}$$

自由空间中的波长

终端加载的无耗传输线



$$V(z) = V_o^+ e^{-j\beta z} + V_o^- e^{j\beta z}.$$

$$I(z) = \frac{V_o^+}{Z_0} e^{-j\beta z} - \frac{V_o^-}{Z_0} e^{j\beta z}.$$

$$Z_L = \frac{V(0)}{I(0)} = \frac{V_o^+ + V_o^-}{V_o^+ - V_o^-} Z_0.$$

反射系数:

$$\Gamma = \frac{V_o^-}{V_o^+} = \frac{Z_L - Z_0}{Z_L + Z_0}.$$

考虑反射系数后的表达式:

$$V(z) = V_o^+ (e^{-j\beta z} + \Gamma e^{j\beta z}),$$

$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} - \Gamma e^{j\beta z}).$$

$$\begin{aligned} \Gamma(-z) &= \Gamma(l) \\ &= \Gamma e^{2j\beta z} \\ &= \Gamma e^{-2j\beta l} \end{aligned}$$

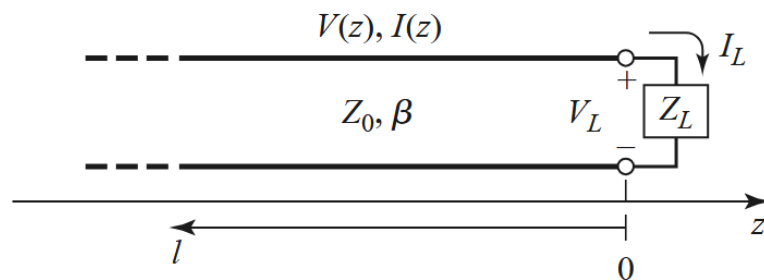
功率关系:

$$P_{\text{avg}} = \frac{1}{2} \text{Re}\{V(z)I(z)^*\} = \frac{1}{2} \frac{|V_o^+|^2}{Z_0} \text{Re}\{1 - \Gamma^* e^{-2j\beta z} + \Gamma e^{2j\beta z} - |\Gamma|^2\} = \frac{1}{2} \frac{|V_o^+|^2}{Z_0} (1 - |\Gamma|^2)$$

这表明在传输线上的任意位置, **平均功率流都是恒定的**。因此, 传递到负载的总平均功率 P_{avg} 等于**入射功率** ($|V_o^+|^2/2Z_0$) **减去反射功率** ($|V_o^+|^2|\Gamma|^2/2Z_0$)。

当 $\Gamma = 0$ 时, **最大功率被传输到负载**; 而当 $|\Gamma| = 1$ 时, **没有功率传输到负载**。

终端加载的无耗传输线



考虑反射系数后的表达式:

$$V(z) = V_o^+ (e^{-j\beta z} + \Gamma e^{j\beta z}),$$

$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} - \Gamma e^{j\beta z}).$$

$$\begin{aligned}
 |V(z)| &= |V_o^+| |1 + \Gamma e^{2j\beta z}| = |V_o^+| |1 + \Gamma e^{-2j\beta \ell}| \\
 &= |V_o^+| |1 + |\Gamma| e^{j(\theta - 2\beta \ell)}|,
 \end{aligned}$$

$$\Gamma = |\Gamma| e^{j\theta}$$

$$V_{\max} = |V_o^+| (1 + |\Gamma|) \quad V_{\min} = |V_o^+| (1 - |\Gamma|)$$

驻波比 (Standing wave ratio, SWR) 或电压驻波比 (Voltage SWR, VSWR)

$$\rho = \text{SWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

反射系数及dB

dB表示为分贝（相对值），通常表示信号强度

$$dB = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{|V_2|^2}{|V_1|^2} = 20 \log_{10} \frac{|V_2|}{|V_1|}$$

反射系数以dB表示 $\Gamma(dB) = 10 \log_{10} |\Gamma|^2$

回波损耗 $RL(dB) = -10 \log_{10} |\Gamma|^2$

反射系数 Γ	回波损耗RL	驻波比VSWR
1.00	0.00	∞
0.90	0.92	19.00
0.80	0.94	9.00
0.70	3.10	5.67
0.60	4.44	4.00
0.50	6.02	3.00
0.40	7.96	2.33
0.30	10.46	1.86
0.20	13.98	1.50
0.10	20.00	1.22
0.09	20.92	1.20
0.08	21.94	1.17
0.07	23.10	1.15
0.06	24.44	1.13
0.05	26.02	1.11
0.04	27.96	1.08
0.03	30.46	1.06
0.02	33.98	1.04
0.01	40.00	1.02
0.00	∞	1.00

- dBm、dBw表示为分贝毫瓦、瓦（相对1mW、1W而言）（功率单位）

$$P_T(dBm) = 10 \log_{10} \frac{P_T}{1mW} \quad P_T(dBW) = 10 \log_{10} \frac{P_T}{1W}$$

知识点

$$a(dB) = 10 \log_{10} a$$

$$2a(dB) = 10 \log_{10} (2a) = 10 \log_{10} 2 + 10 \log_{10} a = \boxed{3(dB)} + a(dB)$$

$$X(dB) + 3dB \rightarrow X * 2$$

$$\text{同理 } X(dB) - 3dB \rightarrow X * 1/2$$

$$a(dB) = 10 \log_{10} a$$

$$10a(dB) = 10 \log_{10} (10a) = 10 \log_{10} 10 + 10 \log_{10} a = \boxed{10(dB)} + a(dB)$$

$$X(dB) + 10dB \rightarrow X * 10$$

$$\text{同理 } X(dB) - 10dB \rightarrow X * 1/10$$

知识点

$$P_T|_{\text{dBm}} = 10 \cdot \log_{10} \left(\frac{P_T}{1 \text{ mW}} \right)$$

$$P_T|_{\text{dBW}} = 10 \cdot \log_{10} \left(\frac{P_T}{1 \text{ W}} \right) = P_T|_{\text{dBm}} - 30 \text{ dB}$$

$$\begin{aligned} \frac{P_1}{P_2} &\triangleq 10 \cdot \log_{10} \left(\frac{P_1}{P_2} \right) \text{ dB} = 10 \cdot \log_{10} \left(\frac{P_1}{1 \text{ W}} \right) - 10 \cdot \log_{10} \left(\frac{P_2}{1 \text{ W}} \right) = P_1|_{\text{dBW}} - P_2|_{\text{dBW}} \\ &= 10 \cdot \log_{10} \left(\frac{P_1}{1 \text{ mW}} \right) - 10 \cdot \log_{10} \left(\frac{P_2}{1 \text{ mW}} \right) = P_1|_{\text{dBm}} - P_2|_{\text{dBm}} \end{aligned}$$

$\text{dBW} \pm \text{dB}$	$= \text{dBW}$
$\text{dBm} \pm \text{dB}$	$= \text{dBm}$
$\text{dBW} - \text{dBW}$	$= \text{dB}$
$\text{dBm} - \text{dBm}$	$= \text{dB}$

$\text{dBm} + \text{dBm}$
 $\text{dBW} + \text{dBW}$


准则:

$$30\text{dBm}=1\text{ W}$$

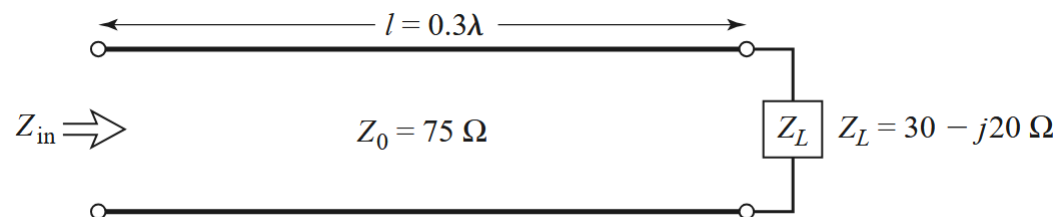
$$X(\text{dBm})+3\text{dB}\rightarrow X*2\text{ mW}$$

$$X(\text{dBm})-3\text{dB}\rightarrow X*1/2\text{ mW}$$

$$X(\text{dBm})+10\text{dB}\rightarrow X*10\text{ mW}$$

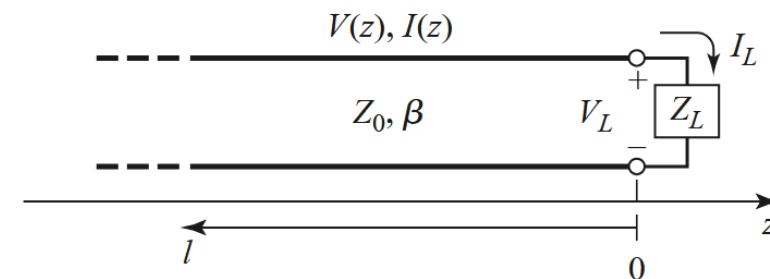
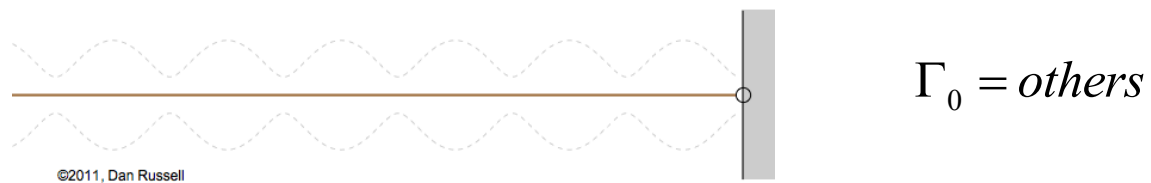
$$X(\text{dBm})-10\text{dB}\rightarrow X*1/10\text{ mW}$$

终端加载的无耗传输线



【例】求传输线上的反射系数分布、输入阻抗 Z_{in} 及电压波分布

终端加载的无耗传输线



$$U(z) = V_0^+ e^{-j\beta z} (1 + \Gamma_0 e^{2j\beta z})$$

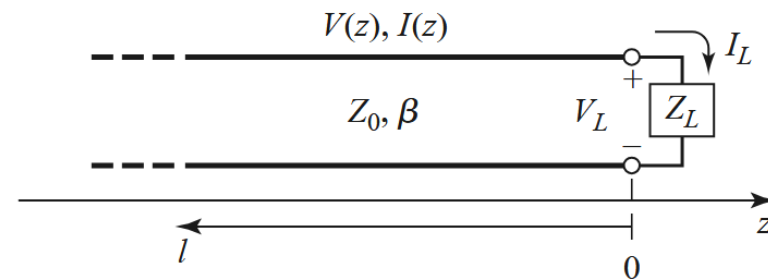
终端加载的无耗传输线

[例] 一根 75Ω 均匀无耗传输线，终端接有负载 $Z_L=R_L+jX_L$ ，欲使线上电压驻波比为3，则负载的实部 R_L 和虚部 X_L 应满足什么关系？

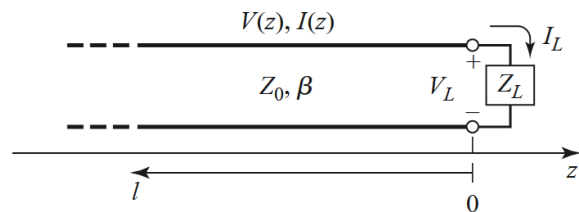
$$|\Gamma_0| = \frac{\rho - 1}{\rho + 1} = 0.5 \quad |\Gamma_0| = \left| \frac{Z_L - Z_0}{Z_L + Z_0} \right| = 0.5$$

$$(R_L - 125)^2 + X_L^2 = 100^2$$

负载的实部 R_L 和虚部 X_L 应在圆心为 $(125, 0)$ 、半径为100的圆上，上半圆对应负载为感抗，而下半圆对应负载为容抗。



终端加载的无耗传输线



$$Z_{in} = \frac{V(-l)}{I(-l)} = \frac{V_o^+ (e^{j\beta l} + \Gamma e^{-j\beta l})}{V_o^+ (e^{j\beta l} - \Gamma e^{-j\beta l})} Z_0 = \frac{1 + \Gamma e^{-2j\beta l}}{1 - \Gamma e^{-2j\beta l}} Z_0$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

具有一定长度的无耗传输线的输入阻抗

$$\begin{aligned}
 Z_{in} &= Z_0 \frac{(Z_L + Z_0)e^{j\beta l} + (Z_L - Z_0)e^{-j\beta l}}{(Z_L + Z_0)e^{j\beta l} - (Z_L - Z_0)e^{-j\beta l}} \\
 &= Z_0 \frac{Z_L \cos \beta l + j Z_0 \sin \beta l}{Z_0 \cos \beta l + j Z_L \sin \beta l} \\
 &= Z_0 \frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l}
 \end{aligned}$$

- $l = \frac{\lambda}{2} \quad Z_{in} = Z_L$

输入阻抗与特征阻抗 Z_0 与传播常数 β 无关

- $l = \frac{\lambda}{4} \quad Z_{in} = Z_0^2 / Z_L$

1. 输入阻抗与与负载阻抗成反比
2. 阻抗变换
3. 短路变开路，开路变短路

- $Z_L = jX_L$

$$Z_{in} = jZ_0 \frac{X_L \cos \beta l + Z_0 \sin \beta l}{Z_0 \cos \beta l - X_L \sin \beta l}$$

纯电抗负载导致输入阻抗也为纯电抗

- $l \ll \lambda, \beta l \approx 0$

$Z_{in} = Z_L$ ，即短线近似

负载短路

$$V(z) = V_o^+ (e^{-j\beta z} - e^{j\beta z}) = -2j V_o^+ \sin \beta z,$$

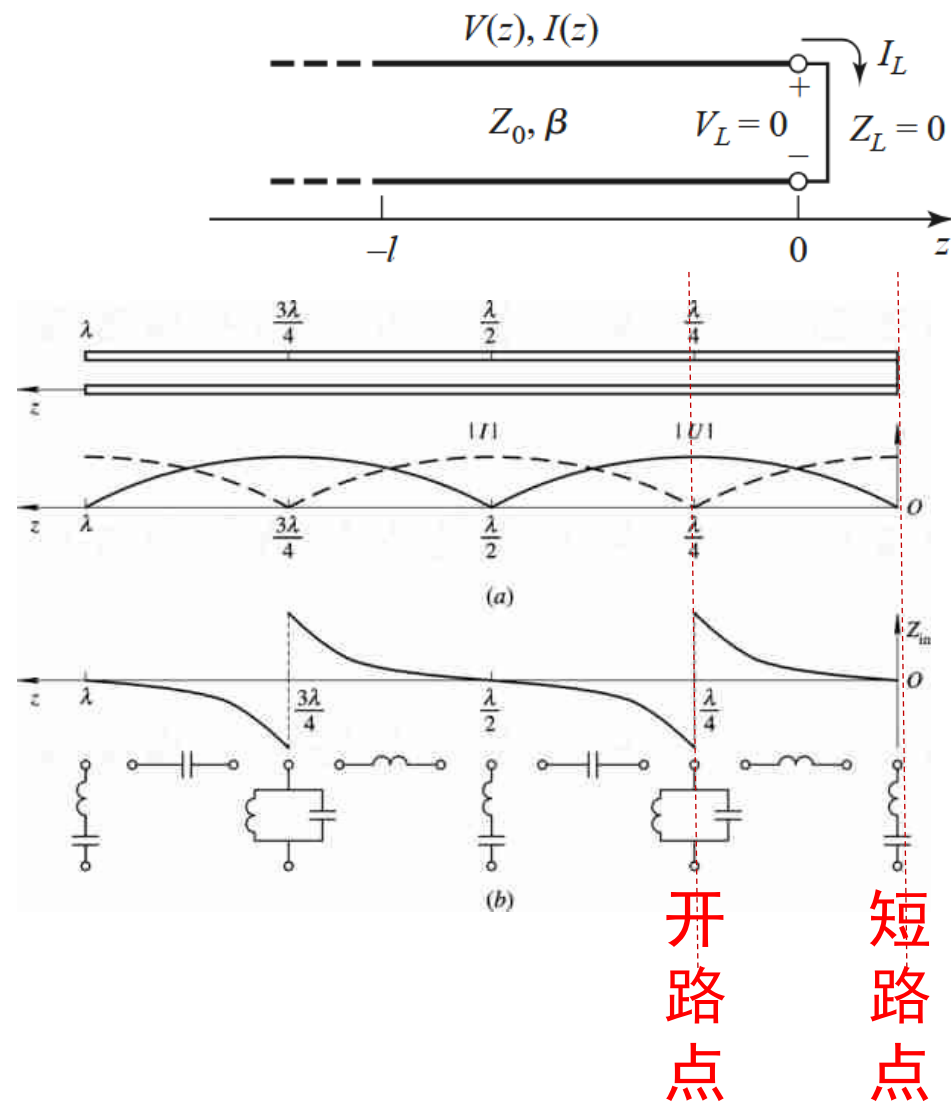
$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} + e^{j\beta z}) = \frac{2V_o^+}{Z_0} \cos \beta z,$$

$$Z_{in} = j Z_0 \tan \beta \ell,$$

$$v(z, t) = \text{Re}\{V(z)e^{j\omega t}\} = 2|V_o^+| \sin \beta z \sin(\omega t + \varphi_0)$$

$$i(z, t) = \text{Re}\{I(z)e^{j\omega t}\} = \frac{2|V_o^+|}{Z_0} \cos \beta z \cos(\omega t + \varphi_0)$$

v/i 相位差了180度



负载开路

$$V(z) = V_o^+ (e^{-j\beta z} + e^{j\beta z}) = 2V_o^+ \cos \beta z,$$

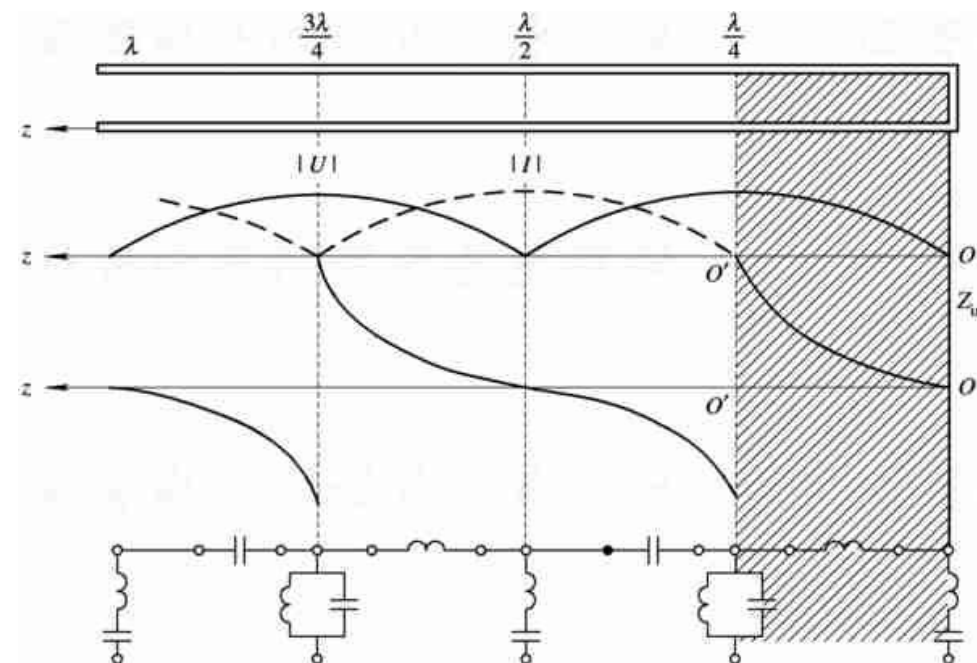
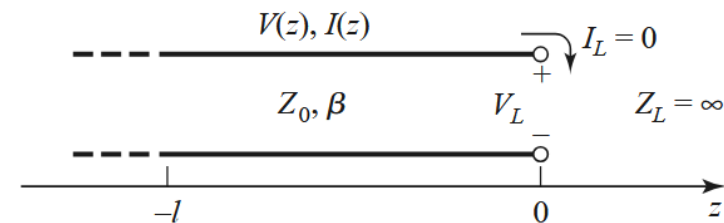
$$I(z) = \frac{V_o^+}{Z_0} (e^{-j\beta z} - e^{j\beta z}) = \frac{-2jV_o^+}{Z_0} \sin \beta z,$$

$$Z_{in} = -jZ_0 \cot \beta \ell,$$

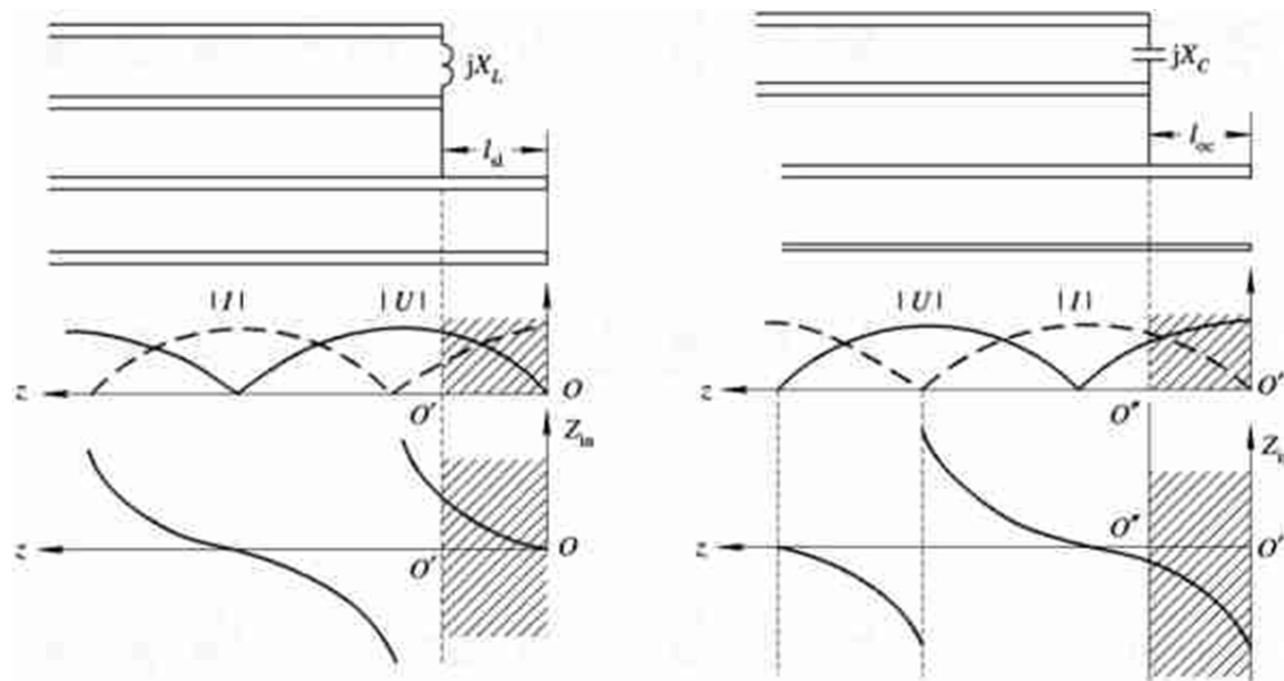
$$v(z, t) = \text{Re}\{V(z)e^{j\omega t}\} = 2|V_o^+| \cos \beta z \cos(\omega t + \varphi_0)$$

$$i(z, t) = \text{Re}\{I(z)e^{j\omega t}\} = \frac{2|V_o^+|}{Z_0} \sin \beta z \sin(\omega t + \varphi_0)$$

v/i 相位差了180度

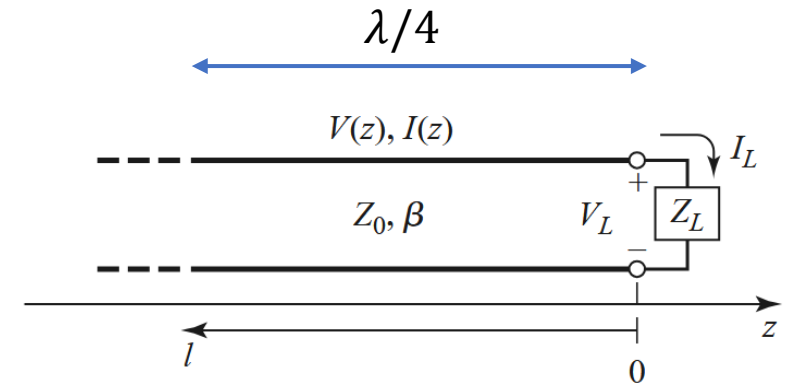


开路/短路状态下的负载特性分析

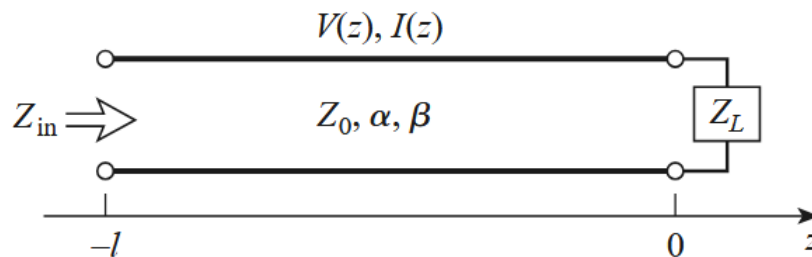


$\lambda/4$ 阻抗变换

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} = \frac{Z_0^2}{Z_L}$$



有耗传输线



损耗因子

$$\gamma = \alpha + j\beta$$

$$V(z) = V_o^+ (e^{-\gamma z} + \Gamma e^{\gamma z}),$$

$$I(z) = \frac{V_o^+}{Z_0} (e^{-\gamma z} - \Gamma e^{\gamma z}),$$

$$Z_{\text{in}} = \frac{V(-\ell)}{I(-\ell)} = Z_0 \frac{Z_L + Z_0 \tanh \gamma \ell}{Z_0 + Z_L \tanh \gamma \ell},$$

$$P_L = \frac{1}{2} \text{Re}\{V(0)I^*(0)\} = \frac{|V_o^+|^2}{2Z_0} (1 - |\Gamma|^2).$$

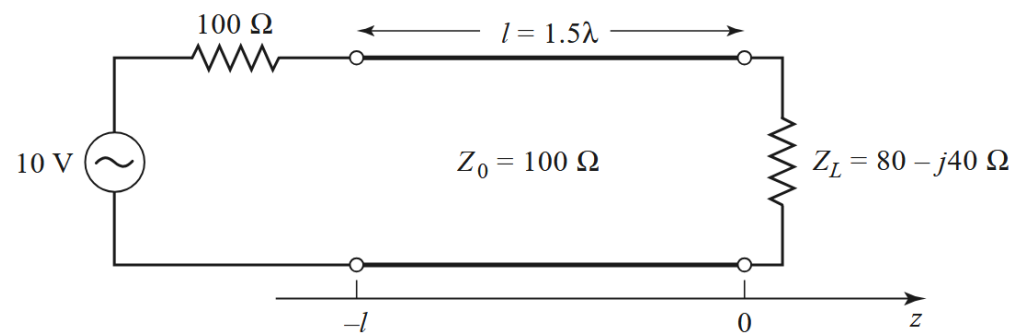
$$P_{\text{in}} = \frac{1}{2} \text{Re}\{V(-\ell)I^*(-\ell)\} = \frac{|V_o^+|^2}{2Z_0} (e^{2\alpha\ell} - |\Gamma|^2 e^{-2\alpha\ell})$$

$$= \frac{|V_o^+|^2}{2Z_0} (1 - |\Gamma(\ell)|^2) e^{2\alpha\ell},$$

$$P_{\text{loss}} = P_{\text{in}} - P_L = \frac{|V_o^+|^2}{2Z_0} [(e^{2\alpha\ell} - 1) + |\Gamma|^2 (1 - e^{-2\alpha\ell})].$$

入射波损耗 反射波损耗

终端加载的无耗传输线



【例】求传输线上的电压波幅值分布

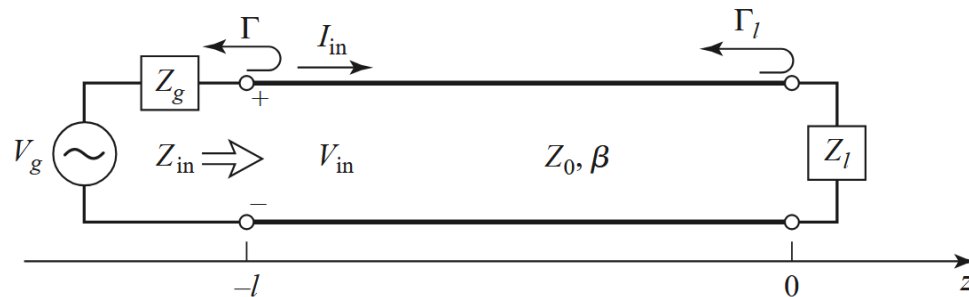
源与负载匹配

$$Z_{\text{in}} = Z_0 \frac{1 + \Gamma_{\ell} e^{-2j\beta\ell}}{1 - \Gamma_{\ell} e^{-2j\beta\ell}} = Z_0 \frac{Z_{\ell} + jZ_0 \tan \beta\ell}{Z_0 + jZ_{\ell} \tan \beta\ell},$$

$$\Gamma_{\ell} = \frac{Z_{\ell} - Z_0}{Z_{\ell} + Z_0}, \quad V(z) = V_o^+ (e^{-j\beta z} + \Gamma_{\ell} e^{j\beta z}),$$

$$V(-\ell) = V_g \frac{Z_{\text{in}}}{Z_{\text{in}} + Z_g} = V_o^+ (e^{j\beta\ell} + \Gamma_{\ell} e^{-j\beta\ell}),$$

$$V_o^+ = V_g \frac{Z_{\text{in}}}{Z_{\text{in}} + Z_g} \frac{1}{(e^{j\beta\ell} + \Gamma_{\ell} e^{-j\beta\ell})} \Rightarrow V_o^+ = V_g \frac{Z_0}{Z_0 + Z_g} \frac{e^{-j\beta\ell}}{(1 - \Gamma_{\ell} \Gamma_g e^{-2j\beta\ell})}, \quad \Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}.$$



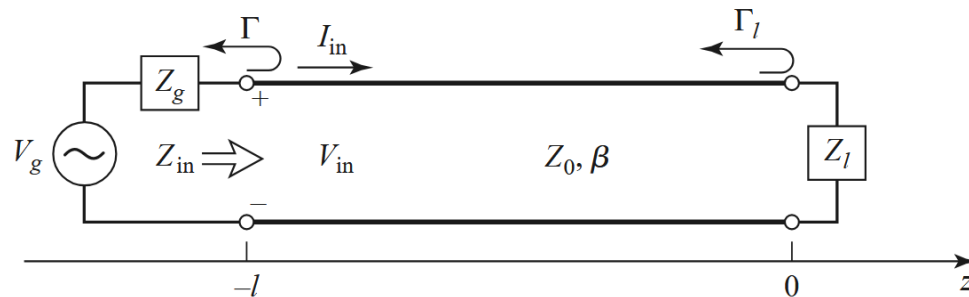
$$P = \frac{1}{2} \text{Re}\{V_{\text{in}} I_{\text{in}}^*\} = \frac{1}{2} |V_{\text{in}}|^2 \text{Re}\left\{\frac{1}{Z_{\text{in}}}\right\} = \frac{1}{2} |V_g|^2 \left|\frac{Z_{\text{in}}}{Z_{\text{in}} + Z_g}\right|^2 \text{Re}\left\{\frac{1}{Z_{\text{in}}}\right\}.$$

$$P = \frac{1}{2} |V_g|^2 \frac{R_{\text{in}}}{(R_{\text{in}} + R_g)^2 + (X_{\text{in}} + X_g)^2}.$$

源与负载匹配

负载与传输线匹配 $Z_L = Z_0$

$$P = \frac{1}{2} |V_g|^2 \frac{Z_0}{(Z_0 + R_g)^2 + X_g^2}$$



源与传输线匹配 $Z_{in} = Z_g$

$$P = \frac{1}{2} |V_g|^2 \frac{R_g}{4(R_g^2 + X_g^2)}$$

$$P = \frac{1}{2} |V_g|^2 \frac{R_{in}}{(R_{in} + R_g)^2 + (X_{in} + X_g)^2}$$

$$\frac{\partial P}{\partial R_{in}} = 0 \rightarrow \frac{1}{(R_{in} + R_g)^2 + (X_{in} + X_g)^2} + \frac{-2R_{in}(R_{in} + R_g)}{[(R_{in} + R_g)^2 + (X_{in} + X_g)^2]^2} = 0, \quad \Rightarrow \quad Z_{in} = Z_g^*$$

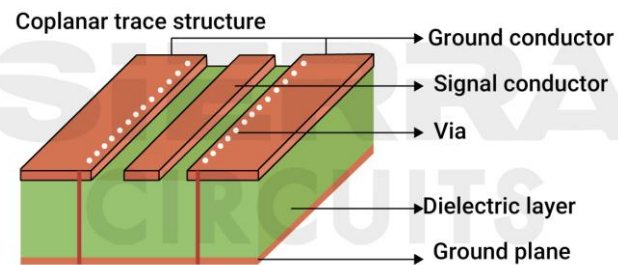
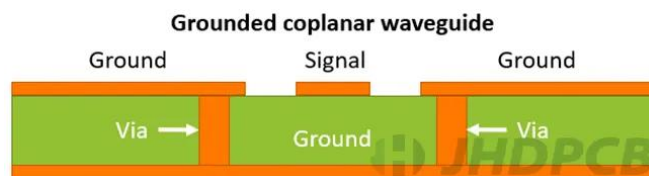
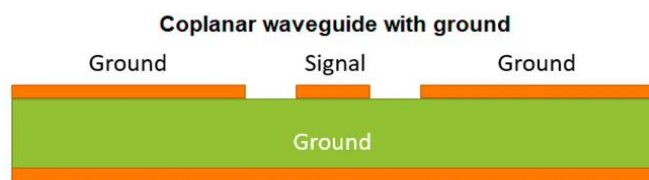
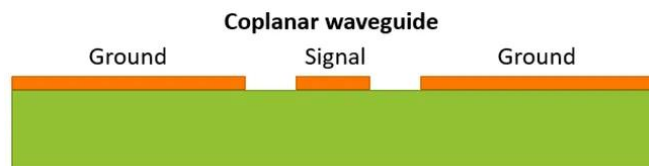
共轭匹配

$$P = \frac{1}{2} |V_g|^2 \frac{1}{4R_g}, \quad \text{但是效率仅有50\%, 若提高效率需尽量降低} Z_g$$

两种情况下的功率谁大谁小?

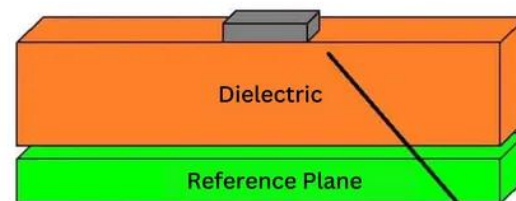
平面传输线

共面波导 Coplanar waveguide, CPW

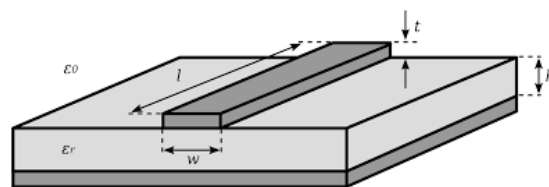


微带传输线 Microstrip line

Microstrip



High-speed transmission line



带状线

Stripline

